PRASHAST VATS

Bangalore, India | +91 63631 40535 | prashastvats@gmail.com | GitHub | Portfolio | Remora (live)

Software Engineer | Full-Stack Product Engineering | Python, TypeScript, React, C#/.NET | AI-Powered Applications, REST APIs, Computer Vision & Cloud Deployment | 3+ Years

KEY SKILLS

- Full-Stack Product Engineering
- REST API Design
- Computer Vision
- React Native
- React & TypeScript
- Relational & NoSQL Data Modelling
- OAuth 2.0 / OIDC
- Cloud Deployment
- Python & FastAPI
- AI-Powered Applications
- Docker & CI/CD
- Cross-Time-Zone Delivery
- C# / .NET MVC
- LLM Integration & Evaluation
- Chrome Extensions (MV3)

PROFESSIONAL EXPERIENCE *Total Experience: 3 years 4 months*

Software Engineer · Mphasis Ltd. (HP Inc. engagement) *May 2023 to Present · 3 yrs 4 mos*

Full-stack engineering, automation, computer vision, and AI tooling for HP's global printing division, working with engineering teams across three time zones.

Product Engineering & AI Tooling

- Designed and built a **.NET MVC web application** end to end, frontend through backend, to scan networked printers over **SNMP**, visualise fleet state, and automate firmware upgrades using **Selenium**. Now integrated into a larger internal platform used by a **10-person team**, letting remote engineers run health checks and firmware upgrades independently instead of routing them through on-site staff.
- Built a computer-vision service using **ResNet** and **FAISS** to identify printer UI elements from screenshots, **eliminating the project's paid Azure AI Vision dependency and its recurring cost entirely**. The service now backs automation flows for **15+ third-party solutions** and runs in **Docker** containers on Ubuntu Linux, on a **$15K to $20K** production server the client commissioned specifically for it.
- Developed an **AI onboarding assistant** using Copilot for the client's automation framework, supporting script explanation, syntax guidance, and code generation. Placed **2nd among 13 developers** in an internal hackathon.

Automation, Delivery & Cross-Team Engineering

- Embedded with HP's **Spain engineering team** on the client's European programme, building automation and MariaDB-backed analytics. The automation **cut project execution time by 15%** and moved testing from office-hours-only manual runs to **continuous 24/7 coverage**. First successful European engagement for Mphasis on this account, and the partnership continues.
- Delivered **5 new features, 5 enhancements to existing features, and 15 bug fixes** across the automation codebase in the last six months.
- Resolved **15+ engineering tickets per month** with approximately **80% first-contact resolution** while supporting **30+ QA engineers** across three time zones.
- Worked across application logic, APIs, database access, network protocols, browser automation, and deployment while debugging production-facing tooling used by HP engineering teams.

CONSULTING EXPERIENCE

Identity Infrastructure Platform · Contributing Engineer Live Demo · Backend · Frontend *2026*

- Built an open-source **delegated-authority system** allowing AI agents to act on behalf of users while maintaining separate agent and user identities.
- Implemented **RFC 8693 token exchange** with short-lived delegation JWTs, **policy middleware**, audit logging, and human approval for destructive operations.
- Built the **React + Tailwind** dashboard for agent configuration, policies, consent approvals, and audit logs.
- Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, React, Vite, Tailwind, Render, Vercel.

BYOC Cloud Infrastructure Platform · Contributing Engineer *March 2026*

Seed-stage BYOC platform spanning AWS, GCP, and Azure, built by a three-person founding team.

- Paired with the platform's frontend engineer to build the company website's public-facing frontend in **React**, **Vite**, **Tailwind CSS**, and **shadcn/ui**.
- Worked with the backend engineer on the **waitlist and interest-capture flow**, persisting prospect contact details to **Supabase** for follow-up on inbound interest.

PROJECTS & INDEPENDENT WORK

Remora, Prompt-Injection Detection for AI Agents remora.watch · Chrome Web Store · npm · Repo *2026 · Published*

- Designed and published a Chrome extension that detects hidden **prompt-injection payloads** targeting AI agents directly in web pages.
- Built the complete browser architecture in **TypeScript**, including **MV3** content scripts, a service worker, a **Shadow DOM** popup UI, the detection engine, and the build pipeline.
- Designed a **dual-signal detection system** combining DOM structure with text-pattern analysis to reduce false positives, achieving **zero false positives** across seven major production sites in a **49-case test suite**.
- Live on the Chrome Web Store at **v0.2.2** with **120+ installs** across three countries, **92% in the United States**. Detection engine also published as a standalone **npm package**, with a dedicated product site at remora.watch.
- Independently shipped the complete product in three weekends, covering engineering, testing, deployment, and Chrome Web Store release.

Autonomous Outbound Voice Agent Live Demo *Winner, Voice Agent Jam 3.0 · May 2026*

- Built and shipped **Priya**, an autonomous voice agent that makes real outbound **SIP/PSTN** calls and handles pricing, competitor, and interest objections in code-switched Hindi/English.
- Designed the **FastAPI** backend to orchestrate Retell AI and Vobiz telephony services and process call outcomes.
- Built a **React** dashboard for enrollment metrics, objection analysis, and per-call sentiment.

- Stack: Python, FastAPI, Retell AI, SIP/PSTN, React, Render.

Event Intelligence Dashboard Live Demo *2026*

- Built an **LLM-backed** application that aggregates multi-source news into continuously updated summaries for user-defined events.
- Designed the backend around **FastAPI** with scheduled background processing using **APScheduler**, and built the React/Vite frontend deployed on Vercel.

Movie Discovery App Live Demo *2026*

- Built a cross-platform mobile movie discovery app in **React Native** with **Expo**, covering search, browsing, and title detail views.
- Implemented a popularity ranking backed by **Appwrite** that tracks search and click activity and surfaces the most-engaged titles as trending.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL

Frontend & Browser: React, React Native, Expo, Vite, Tailwind CSS, shadcn/ui, Angular, HTML5/CSS3, Chrome Extensions, Manifest V3, Shadow DOM

Backend: FastAPI, .NET MVC, ADO.NET, REST API design, middleware, background jobs, webhooks, SNMP

Data: PostgreSQL, Microsoft SQL Server, MariaDB, MongoDB, SQLAlchemy, Supabase, Appwrite

AI & Computer Vision: OpenAI API, Claude API, Gemini API, prompt engineering, LLM evaluation, RAG, FAISS, ResNet, computer vision, voice agents

Auth & Infrastructure: OAuth 2.0, OIDC, JWT, PKCE, Docker, Docker Compose, Jenkins CI/CD, Linux, Render, Vercel, Git, GitHub

EDUCATION & RECOGNITION

B.Tech, Electronics & Instrumentation Engineering · Manipal Institute of Technology *2018 to 2022*

- **Winner, Voice Agent Jam 3.0** (Bangalore, 2026), shipping solo against team entries · **Project Shipyard S2**, an AI Learn Circle cohort supported by Venture Catalysts, Antler, Bhive, and Paytm.
- Certifications: Mphasis TalentNext (React.js, Angular, TypeScript, SQL, Git, GitHub Copilot, Agile) · Interactive Python, UX Design (Coursera).